

In the following exercise, we will perform explanatory data analysis (EDA) to extra insights using Python. Perform the following analyses by continuing from the previous assignment's notebook.

Part A

First, we return to the College dataset. To review, this dataset contains the following variables from 777 different universities and colleges in the US:

- Private : Public/private indicator
 - Apps : Number of applications received
 - Accept : Number of applicants accepted
 - Enroll : Number of new students enrolled
 - Top10perc : New students from top 10% of high school class
 - Top25perc : New students from top 25% of high school class
 - F.Undergrad : Number of full-time undergraduates
 - P.Undergrad : Number of part-time undergraduates
 - Outstate : Out-of-state tuition
 - Room.Board : Room and board costs
 - Books : Estimated book costs
 - Personal : Estimated personal spending
 - PhD : Percent of faculty with Ph.D.'s
 - Terminal : Percent of faculty with terminal degree
 - S.F.Ratio : Student/faculty ratio
 - perc.alumni : Percent of alumni who donate
 - Expend : Instructional expenditure per student
 - Grad.Rate : Graduation rate
1. Use the *seaborn* pairplot() function on the original dataframe with all universities to produce a scatterplot matrix of the first ten columns of the data. Remember to use the *%matplotlib inline* command for plotting in Jupyter. Comment on your observations.
 2. Use the *seaborn* boxplot() function on the original dataframe with all universities to produce side-by-side box plots of 'Outstate' versus 'Private'. **Answer on Canvas: On average, are private universities more expensive than public universities?**
 3. Create a new qualitative variable named *large_university* by binning the 'Enroll' column. We are going to divide universities into two groups based on whether the number of new students enrolled exceeds the average (mean) of all new students enrolled.
 4. Create a dataframe called *large_universities* that only includes large universities.
 5. Use the *pandas* describe(include = 'all') function on *large_universities* to produce a numerical summary of each column. Within this dataframe, what is the 75th percentile for the column 'Enroll'? Name this variable *enroll_answer* and **enter your answer on Canvas.**
 6. Use the *seaborn* boxplot() function on the *large_university* variable to determine whether larger universities are more expensive. **Answer on Canvas: On average, are large universities more expensive than small universities?**

7. To further examine the relationship between the size of a university against its out-of-state tuition, plot a scatter plot. **Answer on Canvas: What type of relationship does this show (weak, medium, strong)?**
8. Use `matplotlib.pyplot.hist()` or `seaborn.distplot()` to produce some histograms with different numbers of bins for a few of the quantitative variables. You may find the `matplotlib.pyplot.subplot()` function useful to plot multiple graphs under a single codeblock. In particular, create a histogram to examine the number of applications received with a bin size of 200. **Answer on Canvas: Are there more universities that receive 0 to 10,000 applications or 10,000 to 20,000 applications?**
9. We are interested in the acceptance rate of these universities. To calculate the acceptance rate, we take the 'Accept' column and divide by the 'Apps' column. Add this column to a copied version of the original dataframe. Remember to copy the dataframe first before performing feature transformation, as we do not wish to alter or inadvertently create new columns in our original data. **Answer on Canvas: Is there a positive or negative relationship between the university's acceptance rate and the percentage of students in the university who graduated from the top 10% from high school?**
10. Continue exploring the data and provide a brief summary of what you discover.

Part B

Next, let's turn to the California housing dataset, which contains information collected from California census districts. The dataset includes the following variables:

- MedInc: median income in block group
- HouseAge: median house age in block group
- AveRooms: average number of rooms per household
- AveBedrms: average number of bedrooms per household
- Population: block group population
- AveOccup: average number of household members
- Latitude: block group latitude
- Longitude: block group longitude
- MedHouseVal: median house value (target variable)

1. To begin, load the California housing dataset. We can fetch this dataset by calling sklearn's API:

```
from sklearn.datasets import fetch_california_housing
import pandas as pd

california_dataset = fetch_california_housing()
california = pd.DataFrame(
    data=california_dataset.data,
    columns=california_dataset.feature_names
)
california['MedHouseVal'] = pd.Series(california_dataset.target)
california.head()
```

2. Make pairwise scatterplots of some predictors (columns) in this dataset. Since this dataset includes many predictors, avoid using the *seaborn* pairplot() function with all the predictors to minimize run-time. Comment on your observations.
3. Are any of the predictors associated with median house value (MedHouseVal)? If so, explain the relationship. **Answer on Canvas: Which predictor has the highest correlation with MedHouseVal, besides MedHouseVal itself?**
4. Do any districts in California appear to have particularly high populations, low median incomes, or unusually high household occupancy rates? Comment on the range of each predictor. Answer on Canvas:
 - a. **Which district index has the highest population?**
 - b. **Which district index has the lowest median income?**
 - c. **Which district index has the highest average household occupancy?**
5. Count the number of districts with latitude between 37 and 38 degrees and longitude between -122 and -121 degrees, corresponding approximately to the San Francisco Bay Area region. Name this number as the variable bay_area_count. Enter your answer on Canvas.

6. Among the districts where median income (MedInc) is greater than 5, what is the median average number of rooms (AveRooms)? Name this number as the variable `median_rooms_high_income`. Enter your answer on Canvas.
7. Among the districts with median income greater than the overall median income, which district has the lowest median house value (MedHouseVal)? Name this index as the variable `min_value_high_income` and enter your answer on Canvas. What are the values of the other predictors for that district, and how do those values compare to the overall ranges for those predictors? Comment on your observations.
8. In this dataset, how many districts have:
 - a. AveRooms greater than the overall median of AveRooms, and
 - b. MedHouseVal greater than the overall median of MedHouseVal?